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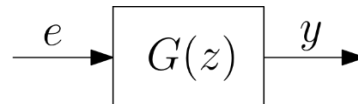
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- =====
- Write the solutions (including procedures and intermediate steps) in the blank areas (use the back of the page, if needed)
 - The number of pages is 4. Additional papers will not be considered.
 - Clarity, order and precision are strongly considered for the final evaluation.
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1. [Analysis of ARMA/ARMAX processes and prediction]

Consider the following stochastic process



where $G(z) = 4z^{-1} + 4z^{-2} + z^{-3}$ and $e(t) \sim WN(1,1)$.

Answer the following questions.

- Explain why the stochastic process is stationary.
- Write the process in canonical form.
- Calculate the process mean value and its covariance function.
- Calculate and draw the power spectral density of the process.
- Write the time domain expression of the 1-step ahead predictor.

Solutions

- The process is stationary since it is the result of an asymptotically stable system $G(z)$ fed by a stationary random noise $e(t)$. The stability of the system can be easily verified computing the poles of the transfer function $G(z)$:

$$G(z) = \frac{4z^2 + 4z + 1}{z^3}$$

All the poles are set in the origin and hence the system is asymptotically stable. Moreover, the process is an MA(3).

- The process $G(z)$ is not written in canonical form. The process can be re-written as follows, leading to

$$y(t) = \frac{4z^2 + 4z + 1}{z^3} e(t) = \frac{4z^2 + 4z + 1}{z^2} e(t) = \frac{z^2 + z + 0.25}{z^2} 4 \cdot e(t) = \frac{(z + 0.5)^2}{z^2} \delta(t) = W(z)\delta(t)$$

where $\delta(t) \sim WN(4, 16)$ and $W(z)$ is the transfer function written in canonical form.

- The mean value is:

$$\mu_y = W(z=1)\mu_\delta = 2.25 \cdot 4 = 9$$

Given that the system is an MA process, the covariance function can be easily computed using the coefficients of the process transfer function:

- $\gamma(0) = (1^2 + 1^2 + 0.25^2) \cdot 16 = 33$
- $\gamma(1) = (1 \cdot 1 + 1 \cdot 0.25) \cdot 16 = 20$
- $\gamma(2) = (1 \cdot 0.25) \cdot 16 = 4$
- $\gamma(3, 4, \dots) = 0$

- d. The power spectral density can be computed using its definition, considering that the non-null covariance function values are only 5. Thus, we have:

$$\Gamma(\omega) = \sum_{\tau=-\infty}^{\tau=\infty} \gamma(\tau) e^{-j\omega\tau} = \gamma(-2)e^{2\omega j} + \gamma(-1)e^{\omega j} + \gamma(0)e^{0\omega j} + \gamma(1)e^{-\omega j} + \gamma(2)e^{-2\omega j}$$

$$\Gamma(\omega) = 33 + 20[e^{\omega j} + e^{-\omega j}] + 4[e^{2\omega j} + e^{-2\omega j}]$$

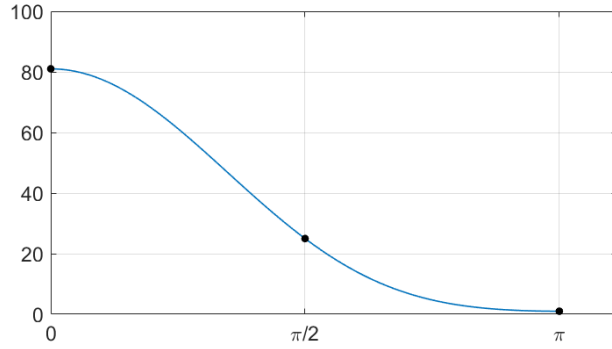
$$\Gamma(\omega) = 33 + 40 \cos(\omega) + 8 \cos(2\omega)$$

To draw the diagram, as the system doesn't have complex poles, we compute its values in the classical three points:

$$\Gamma(\omega = 0) = 33 + 40 + 8 = 81$$

$$\Gamma\left(\omega = \frac{\pi}{2}\right) = 33 - 8 = 25$$

$$\Gamma(\omega = \pi) = 33 - 40 + 8 = 1$$



- e. In order to compute the 1- step ahead predictor the following passages must be done:

1. De-bias the process, since μ_y is not equal to zero

$$\tilde{y} = y - \mu_y \quad \text{and} \quad \hat{\tilde{y}}(t+1|t) = \hat{y}(t+1|t) - \mu_y$$

2. Compute the predictor for the de-biased process

$$\hat{\tilde{y}}(t+1|t) = \frac{C(z) - A(z)}{C(z)} \tilde{y}(t+1) \rightarrow \hat{\tilde{y}}(t+1|t) = \frac{1 + 0.25z^{-1}}{1 + z^{-1} + 0.25z^{-2}} \tilde{y}(t)$$

3. Return to the original process

$$\hat{y}(t+1|t) = \frac{1 + 0.25z^{-1}}{1 + z^{-1} + 0.25z^{-2}} (y(t) - \mu_y) + \mu_y$$

$$\hat{y}(t+1|t) = \frac{1 + 0.25z^{-1}}{1 + z^{-1} + 0.25z^{-2}} y(t) + \left[1 - \frac{1 + 0.25z^{-1}}{1 + z^{-1} + 0.25z^{-2}} \right] \mu_y$$

$$\hat{y}(t+1|t) = \frac{1 + 0.25z^{-1}}{1 + z^{-1} + 0.25z^{-2}} y(t) + \left[\frac{z^{-1} + 0.25z^{-2} - 0.25z^{-1}}{1 + z^{-1} + 0.25z^{-2}} \right] \mu_y$$

That for $t \rightarrow \infty$ becomes:

$$\hat{y}(t+1|t) = \frac{1 + 0.25z^{-1}}{1 + z^{-1} + 0.25z^{-2}} y(t) + \frac{1}{2.25} \mu_y$$

Which in time domain results in the following expression:

$$\hat{y}(t+1|t) = -\hat{y}(t|t-1) - 0.25\hat{y}(t-1|t-2) + y(t) + 0.25y(t-1) + \frac{9}{2.25}$$

2. [Identification/4SID]

Consider a dynamic system where the data of two different experiments are available.

Impulse response data					Step response data				
$t = 0$	$t = 1$	$t = 2$	$t = 3$	$t = 4$	$t = 0$	$t = 1$	$t = 2$	$t = 3$	$t = 4$
$y = 0$	$y = 2$	$y = -1$	$y = 0.5$	$y = -0.25$	$u = 1$	$u = 1$	$u = 1$	$u = 1$	$u = 1$
					$y = 0$	$y = 2$	$y = 1$	$y = 1.5$	$y = 1.25$

The following model classes are also given:

$$\mathcal{M}_1: y(t) = a \cdot y(t-1) + 2u(t-1) + \delta(t)$$

$$\mathcal{M}_2: y(t) = a \cdot y(t-3) + 2u(t-1) + \delta(t)$$

where $\delta(t) \sim WN(0, \lambda^2)$. The parameter $a \in \mathbb{R}$. Answer the following questions.

- Estimate the system order from the impulse response data and select the proper model class $\mathcal{M}$ accordingly.
- Discuss for which values of the parameter a the selected model class can properly represent a stationary stochastic process.
- Using the step response data identify the value of the parameter a .
- Estimate the noise variance λ^2 . Discuss the meaning of the obtained result.

Solutions

- In order to estimate the order of the system, we can use the impulse response data (assuming they are noise free) and compute the Hankel matrix.

$$H_1 = [y(1)] = 2 \rightarrow \text{full rank}$$

$$H_2 = \begin{bmatrix} y(1) & y(2) \\ y(2) & y(3) \end{bmatrix} = \begin{bmatrix} 2 & -1 \\ -1 & 0.5 \end{bmatrix} \rightarrow \det = 0 \rightarrow \text{NOT full rank}$$

So, the system order is $n = 1$. The correct model class is thus $\mathcal{M}_1$ as it features a single pole. $\mathcal{M}_2$, opposite, is a third order system.

- The model can represent a stationary process if it is asymptotically stable, so when $|a| < 1$.
- Firstly, the model class can be written in the transfer function form:

$$y(t) = \frac{B(z)}{A(z)} u(t-d) + \frac{C(z)}{A(z)} \delta(t) \rightarrow y(t) = \frac{2}{1-az^{-1}} u(t-1) + \frac{1}{1-az^{-1}} \delta(t)$$

Recalling the formula for the 1-step ahead predictor for an ARMAX process, it can be concluded that:

$$\hat{y}(t+1|t) = 2 \cdot u(t) + a \cdot y(t)$$

Predictions can be thus computed:

$$\hat{y}(1|0) = 2 + a \cdot 0 = 2$$

$$\hat{y}(2|1) = 2 + a \cdot 2 = 2a + 2$$

$$\hat{y}(3|2) = 2 + a \cdot 1 = a + 2$$

$$\hat{y}(4|3) = 2 + a \cdot 1.5 = 1.5a + 2$$

The cost function to be minimized to find the parameter a is:

$$J_N = \frac{1}{4} \sum_{t=1}^4 (y(t) - \hat{y}(t|t-1))^2$$

$$J_N = \frac{1}{4} [(2-2)^2 + (1-2a-2)^2 + (1.5-a-2)^2 + (1.25-1.5a-2)^2]$$

$$J_N = \frac{1}{4} [(-1-2a)^2 + (-0.5-a)^2 + (-0.75-1.5a)^2]$$

Which is minimized when it holds that:

$$\frac{\partial J_N}{\partial a} = 0 \rightarrow \frac{1}{2} [5.25a + 2.625] = 0 \rightarrow a = \frac{-2.625}{5.25} = -0.5$$

- d. The noise variance can be estimated as the value of the cost function when a is the estimated parameter value. It results:

$$J_N(a = -0.5) = \frac{1}{4} [(-1-2(-0.5))^2 + (-0.5-(-0.5))^2 + (-0.75-1.5(-0.5))^2] = 0$$

The obtained value means that the analyzed process is a deterministic one. In fact, as the noise variance is equal to zero there is no actual stochastic part in the process. This also explains why the determinant of the Hankel matrix is exactly equal to zero, when n is equal or greater of the system order.

3. [ARMA/ARMAX models]

- a) Explain and discuss the structure and features of an all-pass filter
- b) Describe and briefly discuss the canonical representation of an ARMA model

[See lecture notes](#)

4. [ARMA/ARMAX models]

Find (proof requested) the formula for the optimal predictor from data of an ARMAX model.

[See lecture notes](#)